



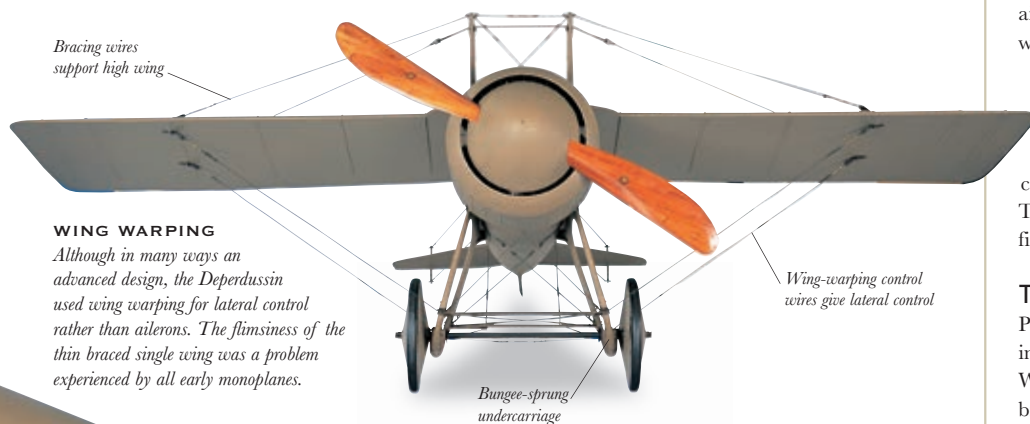
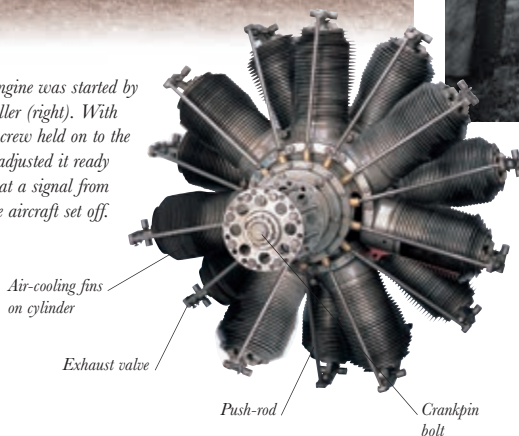
STARTING UP

The Deperdussin's rotary engine was started by manually turning the propeller (right). With the engine running, ground crew held on to the aircraft until the pilot had adjusted it ready for takeoff (above). Then, at a signal from the pilot, they let go and the aircraft set off.



GNOMEPOWER

The Gnome rotary engine had the best power-to-weight ratio of its time. Rotating with the propeller, the engine created torque that pulled sharply to the left, making takeoff tricky.



WING WARPING

Although in many ways an advanced design, the Deperdussin used wing warping for lateral control rather than ailerons. The flimsiness of the thin braced single wing was a problem experienced by all early monoplanes.

STREAMLINED FUSELAGE

The Deperdussin racing plane's monocoque fuselage – a hollow shell of molded plywood – was a milestone in aviation. Monocoque construction would eventually triumph over wooden or metal frame fuselages.

The record-breakers for speed around 1912–13 were the light monoplanes produced by French manufacturers Nieuport, Morane-Saulnier, and above all, Deperdussin (see left), all of which made an attempt at streamlining with a fully enclosed fuselage and engine cowling. In comparison, a biplane such as the Farman Shorthorn, used for military training, was described by a cynical trainee pilot as looking “like an assemblage of birdcages.” But although the monoplanes were sleek and fast, their thin single wing generated inadequate lift for carrying much weight. It was also structurally frail, and was still braced by external wires attached to struts on the fuselage. Their control systems also made these aircraft difficult to handle.

At the time, a thin wing section was considered obligatory by airplane designers. In fact, as aerodynamic research would soon reveal, a thicker wing section provided improved lift, as well as a stronger structure. In 1910 a German high-school professor, Hugo Junkers, took out a patent for “an airplane consisting of one wing, which would house all components, engines, crew, passengers, fuel, and framework.” This flying wing was never built, but the idea led the way to the cantilever wing, requiring no external struts or bracing wires, that Junkers would incorporate into aircraft design during World War I. The cantilever wing would eventually make the monoplane the aircraft of the future. But in 1913–14 the machine that established a new benchmark for performance was a biplane, the Sopwith Tabloid – the first British-designed aircraft to compete successfully for speed with the French. The Tabloid pointed forward to the leading fighter-aircraft design of World War I.

The question of size

Perhaps the greatest technical breakthrough immediately before World War I concerned size. While small, well-designed monoplanes and biplanes were breaking speed and altitude records, to be of any practical use, both in peace and war, airplanes simply had to get bigger. But no one had any clear idea of the feasibility of large flying machines. Greater size implied the use of more than one engine, yet many people doubted that a multiengined aircraft could ever fly safely. Concern centered on how the airplane would behave in the extremely likely event of one of its engines failing. Would this throw the machine into a spin, with fatal consequences? The question was resolved by the young Russian designer and pilot Igor Sikorsky. In 1913–14, he repeatedly flew his large four-engined airplanes – first the Grand and then the Il'ya Muromets – proving that they could remain